Huanqi Wen

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Guangzhou, Guangdong, 510275, China

EDUCATION

- **Sun Yat-sen University** *Guangzhou, China*
  M.S. in Mathematics 09/2023 - Present
  - **GPA:** 89/100; 3.9/4;
  - **Adivisor:** Professor Lei Song
- **Sun Yat-sen University** *Guangzhou, China*
  B.S. in Mathematics, Minnor in Economics 09/2019 - 06/2023
  - **Major GPA:** 89/100; 3.9/4;

RESEARCH INTERESTS

- **Pure mathematics:** toric geometry, syzygies of linear series; tropical geometry.
- **Applied mathematics:** formalization of mathematics, neuroscience and mathematics.

PUBLICATIONS AND PREPRINTS

C=CONFERENCE, J=JOURNAL, P=PATENT, S=IN SUBMISSION, T=THESIS

[S.1] Lei Song, **Huanqi Wen**, Zhixian Zhu. (2024). **On Covering Simplices by Dilations in Dimensions 3 and 4**, arXiv preprint, manuscript submitted for publication in *Discrete & Computational Geometry*, arXiv:2404.02495

RESEARCH EXPERIENCE

- **Positivity of line bundles on toric varieties** *09/2023 - present*
  With Prof. Lei Song
  - **Study** the case where the pair (X, L) of a toric variety X and a line bundle corresponding to a polytope P has the N_p properties.
  - **Theories/topics used:** stability of syzygy bundles, derived category of toric varisties, resolutions of toric subvarieties, Klyachko's classification of toric vector bundles.
  - **Computer tools:** SageMath; Macauley2.
- **Minkowski sum of latitice points in lattice polytopes** *09/2022 - 09/2023*
  With Prof. Lei Song and Prof. Zhixian Zhu
  - **Studied** which lattice polytopes are normal, specifically whether the lattice points in an $n+1$ dilation of a polytope P can be written as the sum of a lattice point in P and a lattice point in an n dilation of P.
  - **Original problem:** Oda's conjecture.
  - **Results:** by covering simplices with dilations, I proved that simplices with large lattice length in dimensions 3 and 4 are normal.
  - **Wrote** computer program (Python, matlab) to evaluate the normality of lattice polytopes with large lattice length.
- **Tropical geometry** *09/2023 - 01/2024*
  Explored the problem "Curve Counting Problems and Tropical Geometry" and wrote a survey for a graduate-level course.

READING SEMINAR

- **Speaker - Toric Geometry Seminar** *2024 Spring - 2024 Fall*
  Fulton's book "Introduction to Toric Varieties" and other classical papers [website]
- **Speaker - Syzigies Seminar** *2025 Spring*
  Irena Peeva's book "Graded Syzygies"
- **Algebraic geometry Seminar** *2024 Spring - 2025 Spring*
  Hartshorne's book "Algebraic Geometry"

CONFERENCES

- **2024 International Summer school of Algebraic Geometry** *07/2024*
  Fudan University [website]
- **China Southeastern Algebraic Geometry Symposium VII** *12/2024*
  Sun Yat-sen University [website]
- **Stable reduction of algebraic curves** *05/2025*
  Xiamen University, Speaker: Qing Liu
- **2025 SRI in algebraic geometry** *07/2025*
  Colorado state university [website]

EXPERIENCE

- **Sun Yat-sen university** *2025 Fall*
  Graduate teaching assistant of linear algebra.
- **International Scientific Research and Development Training** *2025 Spring*
  Completed the course on "Constructive Mathematics and Theoretical Computer Science," taught by Professor Vladimir Cherno.
- **Guangdong Digital Talent Training Program** *07/2022 - 08/2022*
  Conducted practical training and internships at a government digitalization department and China Telecom.
- **Department Student Union Member and Head of a Division** *09/2019 - 06/2021*
  Planned and led diverse student activities within the department

AWARDS AND SCHOLARSHIPS

- **Graduate Scholarship (Top 20 %)** *2023-2024, 2024-2025,2025-2026*
  Sun Yat-sen University
- **Honorable Mention, Mathematical Contest in Modeling (MCM)** *2022*
  Organized by the Consortium for Mathematics and Its Applications (COMAP)
  - Awarded for outstanding performance in solving a real-world mathematical modeling problem as part of a three-person team in an international modeling competition.
- **Postgraduate Scholarship (Top 20 %)** *2020-2023, three times*
  Sun Yat-sen University

SKILLS

- Languages: English (IELTS: 7.0)
- Programming Languages: C++, Python
- Mathematical tools: MATLAB, SageMath, Macaulay2, Latex